

Text Styles and Layouts**I Practical Significance** *(with respect to practicum no. 4)*

Employing CSS text styles and layouts for your webpage design transcends mere aesthetics; it's a practical necessity for effective communication and user engagement. Well-defined text hierarchy, legible fonts, and intuitive layouts ensure that visitors can effortlessly scan and comprehend crucial information, whether they're seeking local businesses, accessing online services, or engaging with content on their mobile devices. This clarity enhances user experience, reduces bounce rates, and ultimately contributes to achieving the webpage's objectives, be it driving conversions, disseminating information, or building a strong online presence within the local community.

II Competency and Industry-Specific Practical Skills *(Kindly copy from the Course Structure)* This practicum exercise is expected to develop the following skills for the industry-identified competency.

- **Develop** interactive web applications.

a) **Apply** CSS for visual styling and effective layout of web content. *(Psychomotor Domain Outcome)*

b) **Practice** safe coding and good privacy habits. *(Affective Domain Outcome)*

c) **Adhere** to laboratory safety protocols. *(Affective Domain Outcome)* d) **Interpret** the observations and results. *(Cognitive Domain Outcome)*

III Relevant CO(s) *(Kindly copy from the Course Structure)*

CO2. Use CSS styles to format and layout static web pages.

IV Practical Outcome (Pro) *(Kindly copy from the Course Structure)*

- **Use** text styles and layouts for the given webpage design.

V Related ADO(s) *(Kindly copy from the Course Structure)*

a) **Follow** ethical coding standards and responsibilities.

b) **Practice** persistence in styled designs.

c) **Adopt** a self-learning approach to web frameworks.

d) **Engage** and cooperate in collaborative coding projects.

VI Minimum Underpinning Theory *(Theory should be written only to relate this practicum)*

Minimum justification for employing CSS text styles and layouts in webpage design boils down to basic usability and information conveyance. Without structured text and a discernible layout, content devolves into an unreadable jumble, frustrating users and rendering the webpage functionally useless for communicating its intended message, be it local business details or essential service information. Even the most rudimentary application of CSS ensures a minimal level of organization, allowing visitors to at least differentiate headings from body text and navigate the content, thereby preventing immediate abandonment and offering a sliver of opportunity for engagement within the local online context.

VII Practicum Set-up/Circuit Diagram/Algorithm/Flowchart**VIII Resources Required**

S.No.	Name of Resource	Suggested Broad Specification	Quantity
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1	Hardware	Minimum Intel Core i5/i7 with 8GB RAM and 256GB SSD, Linux-based or Windows-based machines	1 No.
2	Text Editor	Microsoft Visual studio (VS Code), Version-1.68	1 No.

IX Safety Precautions

- Keep** Liquids Away from Equipment:
- No** or Snacks
- Use** Equipment Properly
- Maintain** Proper Posture
- Avoid** Overloading Power Outlets
- Follow** institutional laboratory policies.
- Avoid** storing confidential data in lab computers.
- Use** equipment only for academic purposes.

X Procedure

- Create** project folder structure
- Set up** the HTML skeleton
- Style** the layout
- Add** visual enhancements
- Test** responsiveness

XI Actual Resources Used *(To be filled by the students)*

S.No.	Name of Resource	Broad Specification	Quantity	Remarks(if any)
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1	Operating System	Windows 11	1 No.	System running smoothly
2	Programming Language	HTML5	1 No.	Standard
3	Software Development Kit	VS Code (Version 1.135.0)	1 No.	Text Editor
4	Libraries	N/A	N/A	Not Used
5	Graphics/Animation tool	N/A	N/A	Not Used
6	Front-end tool	Google Chrome	1 No.	Web Browser
7	Back-end tool	N/A	N/A	Not used
8	Virtual Lab	N/A	N/A	Not used
9	Specific Hardware	Intel Core i5, 16GB RAM	1 No.	Personal Laptop
10	Simulation/Emulation Tool	N/A	N/A	Not used

XII Actual Procedure Followed *(To be filled by the students on the left page)*

- Designed and structured the web page layout using semantic HTML tags (<header>, <aside>, <main>, <h1>, <h2>, <p>).

2. Applied CSS styling to set up typography using Google Fonts (Roboto), define color themes, and configure element spacing/borders.
3. Implemented a two-column layout using CSS Flexbox (`display: flex`) with a 1:2 proportion between the sidebar and main content areas.

XIII Observations *(To be filled by the students on the left page if space is not sufficient)*

1. The CSS Flexbox property successfully arranged `.sidebar` and `.main-content` horizontally side-by-side within `.layout-container`.
2. Setting `text-indent: 20px` on paragraph tags (`p`) indented the body text, while explicitly setting `text-indent: 0` on `.price` prevented unintended paragraph indentation inside the pricing card.
3. Using custom properties like `letter-spacing`, `line-height`, and explicit text colors enhanced visual hierarchy and typography readability.

XIV Results *(To be filled by the students)*

1. Successfully created a responsive two-column web layout for "SkillBridge" displaying skill domain categories and mentorship details.
2. Formatted a distinct pricing card component with custom typography, rounded borders, and background contrast.
3. Corrected non-standard CSS properties (e.g., replacing `min-font-size` with `font-size`) to ensure proper cross-browser rendering.

XV Interpretation of Results *(Students to state the meaning of the above-obtained results/observations)*

The implementation demonstrates how HTML5 semantic elements paired with CSS Flexbox create structured, clean, and modern web interfaces. By separating page layout into flex containers and applying targeted typography rules, complex visual components like pricing cards and sidebars can be structured predictably and maintained efficiently.

XVI Conclusions *(Students to draw conclusions (or take decisions) based on the interpretation of results).*

The implementation of the SkillBridge web interface successfully demonstrates the core principles of modern, semantic web layout and typographic design. By structuring the document with semantic HTML elements such as `<header>`, `<aside>`, and `<main>`, the page achieves a clear structural hierarchy and optimal accessibility. The use of CSS Flexbox enables a flexible and maintainable two-column layout, seamlessly dividing content between the navigation sidebar and the main content area in an ideal 1:2 ratio. Furthermore, applying deliberate styling rules—including defined font hierarchies with Google Fonts, custom spacing, reset indents, and distinct component styling for elements like pricing cards—significantly enhances visual clarity and readability. Overall, this practicum highlights how HTML5 structure paired with CSS flexbox layout techniques leads to scalable, efficient, and user-friendly web interface designs.

XVII Suggested Practicum Related Questions:

Note: Below given are a few sample questions for reference. Teachers must design more such questions in order to ensure the achievement of pre-determined course outcome. 1) How would you set the font family to 'Arial' and fallback to sans-serif using CSS? 2) Write CSS to make all `<h2>` elements bold and 24px in size.

3) How do you center-align text in a `<div>` and set its color to dark blue

(#00008B)? 4) How would you transform all text inside `<span>` to uppercase using CSS?

5) How to Set the font style of `<em>` to italic and increase letter spacing by 1px? 6) How to Set line height to 1.5 and word spacing to 3px for paragraph text. 7) How do you indent the first line of a paragraph by 20px?

8) How would you import 'Roboto' from Google Fonts and apply it to `<body>`? 9) How would you ensure a minimum font size of 16px even on small screens?

XVIII References / Suggestions for further reading

1. **Books:** Kumar, Rawat Tarun , 'Web Technology', Oxford University Press, New Delhi, 2015, ISBN: 9780198081937

2. **Web References:** <https://www.w3schools.com/css/>

3. **Video Links:** Introduction to Web Technology – NPTEL Video - https://www.youtube.com/watch?v=6dFnpz_AEYA

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product related marks.

Note: The weightage for the **process-related skills** and **product-related skills** will change depending upon the PrO, COs, and competency related to the concerned course.

Performance Indicators	Weightage	Marks Obtained
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Process related skills: 18 Marks - 60%		Weightage	Marks Obtained
1	Understanding Design Principles	10%	
2	Font Styling	20%	
3	Text Alignment and Decoration	20%	
4	Responsiveness and Accessibility	10%	
Product related skills: 12 Marks - 40%			
5	Effective Product Description	10%	
6	Text Size and Layout Adjustments	10%	
7	Text Styling for Product Pricing	05%	
8	Conclusions	05%	
9	Oral viva examination	05%	
10	Submission of code and report on time	05%	
	Total	100%	

Names of Student Team Members with Register number:

1.
2.
3.
4.

Marks Obtained			Dated Signature of Teacher
Process related Skills (18)	Product related skills (12)	Total (30)	